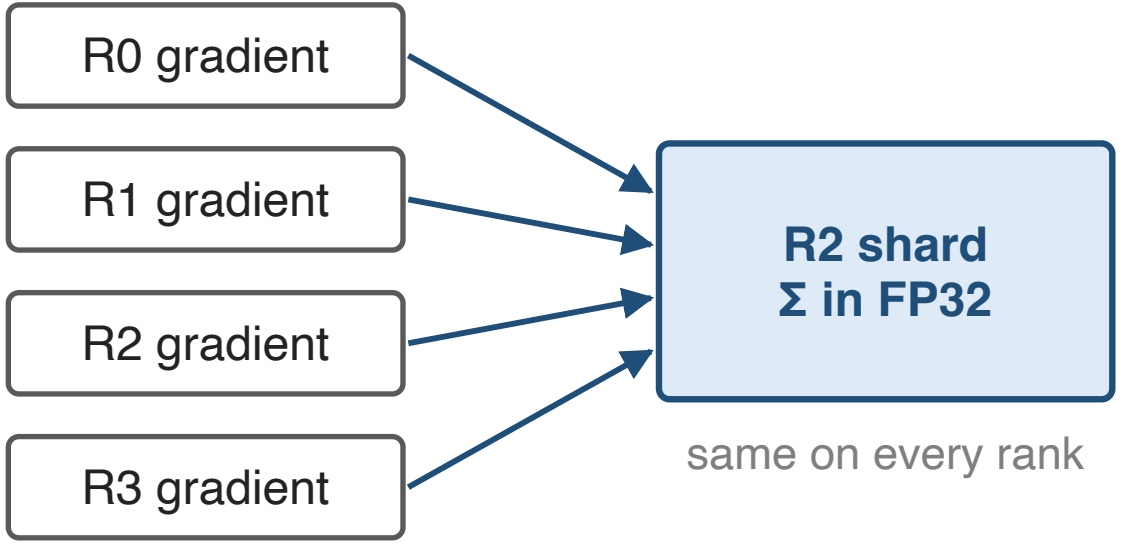
(a) Ring reduce-scatter in BF16

rounding error at every hop

(b) V4: all-to-all, then FP32 sum

BF16 with stochastic rounding



one rounding, FP32 accumulation